

Propiedades de una derivación

Lema 1. Sean $p \in M$ variedad diferenciable, $w \in T_p(M)$ y $f, g \in C^\infty(\mathbb{R}^n)$, entonces

$$(1) \ f \text{ constante} \implies w(f) = 0. \qquad (2) \ f(p) = g(p) = 0 \implies w(fg) = 0.$$

Demostración:

(1) Basta probarlo para $f_1 \equiv 1$ ya que para $f \equiv c \in \mathbb{R}$ se tiene por linealidad que $w(f) = w(cf_1) = cw(f_1) = 0$. Para f_1 , podemos aplicar la regla de Leibniz:

$$w(f_1) = w(f_1 \cdot f_1) = f_1(p)w(f_1) + f_1(p)w(f_1) = 2w(f_1) \implies w(f_1) = 0.$$

(2) De manera similar al apartado anterior, podemos aplicar la regla de Leibniz:

$$w(fg) = f(p)w(g) + g(p)w(f) = 0 + 0 = 0. \quad \blacksquare$$

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